

Cybersecurity Internship Report: Task 1

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Objective:

Discover open ports on local and remote devices to evaluate network exposure and identify potential security risks.

Tools I have Used:

1. Nmap - Network discovery and scanning tool.
2. Wireshark - Network packet analysis tool.
3. Gobuster - Web directory brute-forcing tool.

Network Reconnaissance & Findings

performed port scans using **Nmap** on both my local virtual environment and a web called testphp.vulnweb.com where testing is allowed.

A. Local Network Scan

As shown in my terminal screenshot, scanning the subnet revealed several active hosts:

```
File Actions Edit View Help
(FebinMaxon@kali)-[~]
$ nmap -sS 192.168.162.132/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-12 08:41 EST
Nmap scan report for 192.168.162.1
Host is up (0.00041s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:50:56:C0:00:08 (VMware)

Nmap scan report for 192.168.162.2
Host is up (0.00026s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    filtered domain
MAC Address: 00:50:56:FD:28:1C (VMware)

Nmap scan report for 192.168.162.254
Host is up (0.00052s latency).
All 1000 scanned ports on 192.168.162.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:FB:21:AD (VMware)

Nmap scan report for 192.168.162.132
Host is up (0.00013s latency).
All 1000 scanned ports on 192.168.162.132 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

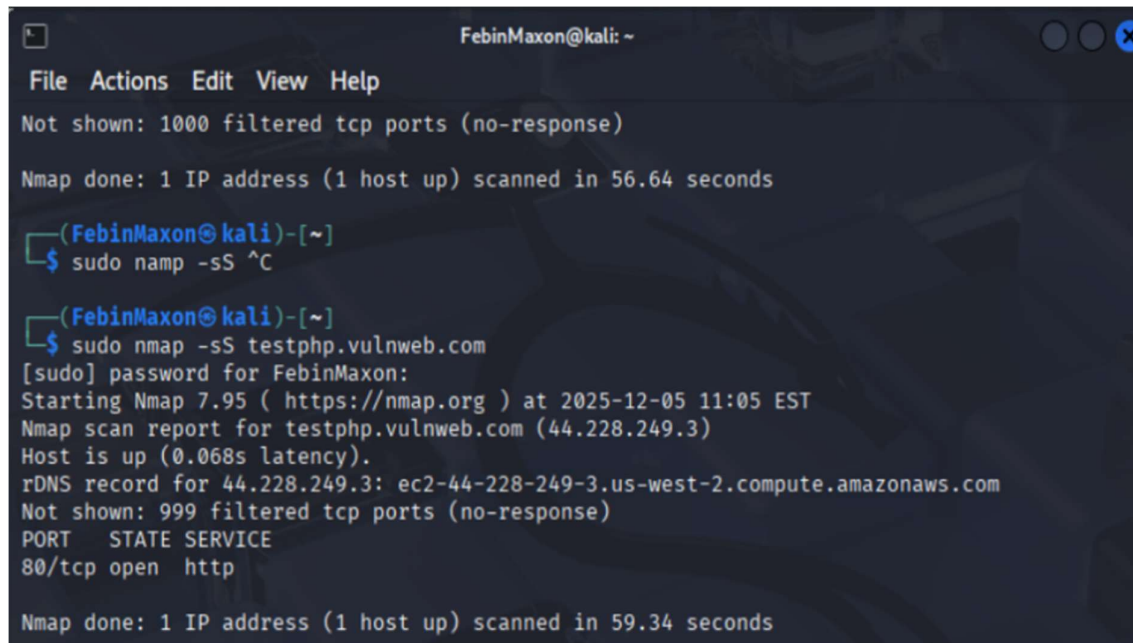
Nmap done: 256 IP addresses (4 hosts up) scanned in 41.50 seconds
```

- **Target \$192.168.162.1\$ (VMware Host):**
- **Port 139/tcp (netbios-ssn):** NetBIOS Session Service.
(Risk Level: Medium, Enables NetBIOS enumeration (usernames, shares, system info), Used in credential harvesting attacks in poorly secured networks.)
- **Port 445/tcp (microsoft-ds):** SMB (Server Message Block) used for file/printer sharing.
(Risk Level: High, Exploitable vulnerabilities (e.g., EternalBlue / WannaCry), Can allow remote execution if misconfigured)

- **Port 53/tcp (domain):** Filtered; likely a DNS service.
(Risk Level: Low–Medium or lower if properly firewalled, If misconfigured, can allow DNS zone transfer attacks, Filtered status suggests firewall protection, reducing risk)

B. Improvised Scan on testphp.vulnweb.com

I extended the task by scanning testphp.vulnweb.com



```

FebinMaxon@kali: ~
File Actions Edit View Help
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 56.64 seconds

(FebinMaxon@kali)-[~]
$ sudo nmap -sS ^C

(FebinMaxon@kali)-[~]
$ sudo nmap -sS testphp.vulnweb.com
[sudo] password for FebinMaxon:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-05 11:05 EST
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.068s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 59.34 seconds

```

- **Port 80/tcp (http):** The default port for unencrypted web traffic.
(Risk Level: Medium–High)
 - depends on whether HTTPS is enforced and web app security.
 - Data transmitted is **unencrypted**, making it vulnerable to **packet sniffing** (credentials, session cookies, form data).

- Susceptible to **Man-in-the-Middle (MITM)** attacks.
- Vulnerable to common web attacks if the application is insecure (e.g., **SQL Injection, XSS, file inclusion**).

C. Directory Enumeration Findings (Gobuster)

I Used directory brute-forcing, paths were identified:

```

FebinMaxon@kali: ~
File Actions Edit View Help

(FebinMaxon@kali)~$ gobuster dir -u http://testphp.vulnweb.com/ -w /usr/share/seclists/Discovery/Web-Content/common.txt -x php,html,txt -t

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://testphp.vulnweb.com/
[+] Method: GET
[+] Threads: 50
[+] Wordlist: /usr/share/seclists/Discovery/Web-Content/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: php,html,txt
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

/404.php (Status: 200) [Size: 5267]
/ CVS (Status: 301) [Size: 169] [→ http://testphp.vulnweb.com/ CVS/]
/ CVS/Entries (Status: 200) [Size: 1]
/ CVS/Repository (Status: 200) [Size: 8]
/ CVS/Root (Status: 200) [Size: 1]
/ admin (Status: 301) [Size: 169] [→ http://testphp.vulnweb.com/ admin/]
/ artists.php (Status: 200) [Size: 5328]
/ cart.php (Status: 200) [Size: 4903]
/ categories.php (Status: 200) [Size: 6115]
/ cgi-bin (Status: 403) [Size: 276]
/ cgi-bin.html (Status: 403) [Size: 276]
/ cgi-bin/ (Status: 403) [Size: 276]
/ cgi-bin/.html (Status: 403) [Size: 276]
/ crossdomain.xml (Status: 200) [Size: 224]
/ disclaimer.php (Status: 200) [Size: 5524]
/ favicon.ico (Status: 200) [Size: 894]
/ guestbook.php (Status: 200) [Size: 5390]
/ images (Status: 301) [Size: 169] [→ http://testphp.vulnweb.com/ images/]
/ index.php (Status: 200) [Size: 4958]
/ index.php (Status: 200) [Size: 4958]
/ login.php (Status: 200) [Size: 5523]
/ logout.php (Status: 200) [Size: 4830]
/ pictures (Status: 301) [Size: 169] [→ http://testphp.vulnweb.com/ pictures/]
/ product.php (Status: 200) [Size: 5056]
/ redirect.php (Status: 302) [Size: 0]
/ search.php (Status: 200) [Size: 4732]
/ secured (Status: 301) [Size: 169] [→ http://testphp.vulnweb.com/ secured/]
/ signup.php (Status: 200) [Size: 6033]
/ userinfo.php (Status: 302) [Size: 14] [→ login.php]
/ vendor (Status: 301) [Size: 169] [→ http://testphp.vulnweb.com/ vendor/]
Progress: 19000 / 19000 (100.00%)

Finished

```

/admin

Risks:

- Likely administrative panel.

- High risk of **brute-force attacks** if authentication is weak.
- May expose sensitive configuration or management functions.
- If improperly secured, could allow **privilege escalation**.

Risk Level: High

/login.php

Risks:

- Target for **credential brute-forcing** and password spraying.
- Vulnerable to **SQL injection** if input validation is weak.
- May expose user enumeration through different error messages.
- Risk of session hijacking if not protected by HTTPS.

Risk Level: Medium–High

/secured/

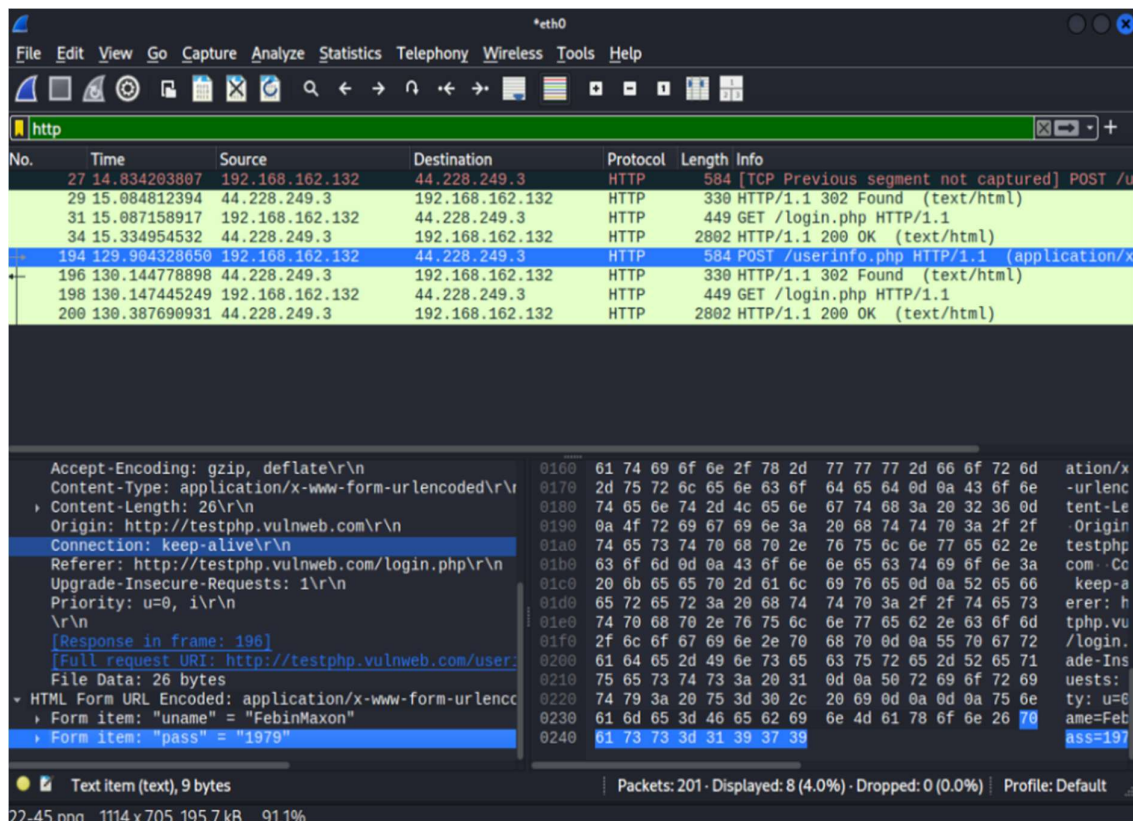
Risks:

- Suggests restricted or sensitive content.
- If access controls are misconfigured, may allow **unauthorized data access**.
- Possible exposure of confidential files or backups.
- May be vulnerable to **directory listing** if indexing is enabled.

Risk Level: Medium–High (depends on access control implementation)

D. Wireshark Traffic Analysis – HTTP Credential Exposure

- HTTP **POST request** sent to /userinfo.php
- Content-Type: application/x-www-form-urlencoded
- Credentials visible in plaintext:
- uname = FebinMaxon
- pass = 1979
- Connection over **HTTP (not HTTPS)**



Security Risks

1. Plaintext Credential Exposure (Critical Risk)

- Username and password are transmitted **unencrypted**.
- Anyone on the same network can capture credentials using packet sniffing tools (e.g., Wireshark).
- Enables **credential theft and account compromise**.

Risk Level: Critical

2. Lack of HTTPS Encryption

- No TLS/SSL protection.
- Vulnerable to:
 - **Man-in-the-Middle (MITM) attacks**
 - Session hijacking

- Traffic manipulation
- Attackers can modify or replay requests.

Risk Level: High

3. Session & Data Integrity Risk

- Login responses (302 redirect → 200 OK) also unencrypted.
- Session cookies (if present) could be stolen.
- Enables **unauthorized access without password cracking**.

Risk Level: High

Conclusion

This task provided a hands-on introduction to the foundational phase of any security assessment: **Network Reconnaissance**. By utilizing **Nmap**, I successfully identified active hosts and open services within both a local virtual environment and a remote target.